

WiFi-CSI Cross-Environment Calibration

Why sensing collapses across environments and how few-shot calibration recovers it, measured

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Abstract

WiFi channel state information (CSI) sensing is attractive because a single router can detect presence, falls, or motion without any dedicated sensor. The question that actually matters in deployment, though, is not ‘what accuracy does it reach’ but ‘how much of that accuracy survives once the device leaves the room it was trained in.’ CSI amplitude is strongly shaped by the receiver’s automatic gain control, antenna placement, and the multipath reflections created by walls and furniture, so a classifier trained in one physical condition is widely believed to degrade sharply in another. This paper measures that belief directly, reports why the measurement turned out to be harder than expected, and quantifies how much of the resulting loss a small amount of target-domain calibration data can recover.

Contents

1	Abstract	3
2	The Problem: Why WiFi-CSI Breaks When the Environment Changes	3
3	Method: Domain Splits and Zero-Shot vs. Few-Shot Calibration	3
4	Data and Setup: What Split Was Actually Available	4
5	Results	5
6	Limitations	7
7	Reproduction	7
8	References	8

1 Abstract

WiFi channel state information (CSI) sensing is attractive because a single router can detect presence, falls, or motion without any dedicated sensor. The question that actually matters in deployment, though, is not “what accuracy does it reach” but “how much of that accuracy survives once the device leaves the room it was trained in.” CSI amplitude is strongly shaped by the receiver’s automatic gain control, antenna placement, and the multipath reflections created by walls and furniture, so a classifier trained in one physical condition is widely believed to degrade sharply in another. This paper measures that belief directly, reports why the measurement turned out to be harder than expected, and quantifies how much of the resulting loss a small amount of target-domain calibration data can recover.

The experiment runs on the public UT-HAR dataset (SenseFi redistribution, 7 activities, 250-step by 90-channel CSI amplitude windows) through a classical feature-extraction (158 handcrafted dimensions) and LightGBM 7-way activity classification pipeline. The first domain-shift axis we tried was the only grouping variable this redistribution actually publishes: the official train/val/test boundary. It showed essentially no domain shift at all (zero-shot accuracy 94.7-94.9%, matching the in-split baseline of 95.2% within noise), which is exactly what the SenseFi paper’s own caveat about sliding-window overlap causing leakage across that boundary predicts. Because the published metadata cannot support a verified cross-room, cross-subject, or cross-day split, we constructed our own domain-shift proxy with unsupervised clustering on each window’s per-channel mean amplitude, a physical fingerprint of receiver gain and multipath. On this constructed split (“pseudo-session”), the story flips completely. A classifier trained on the source domain (3624 windows) and applied zero-shot to the target domain (1349 windows) collapses to 55.08% accuracy. Training on target-domain data alone caps out at 92.37%, so the domain-shift gap is 37.29 percentage points. Adding 1, 5, 10, 20, or 50 labeled target examples per class to the source training set recovers 57.16%, 64.55%, 70.62%, 78.23%, and 84.92% accuracy respectively, which is 5.6%, 25.4%, 41.7%, 62.1%, and 80.0% of that gap. Adding just 350 labeled target-domain examples in total, 50 per class, only 9.7% of the source training set’s size, recovers 80% of the gap.

2 The Problem: Why WiFi-CSI Breaks When the Environment Changes

WiFi-CSI-based presence and activity sensing has been an active research topic for over a decade, appealing precisely because it needs no camera or wearable, only the physical-layer signal from a router already installed in the home. The catch is that CSI amplitude responds not just to the activity but to the physical environment the activity happens in. The same person doing the same motion can produce a completely different amplitude scale and fluctuation structure depending on the receiver’s automatic gain control (AGC) setting, the distance between antenna and body, and the multipath reflection pattern created by room size and furniture layout. A classifier trained in one environment is repeatedly reported to degrade sharply when moved to another, and this is one of the central obstacles to real deployment. It is also why a line of work including Widar3.0 (Zheng et al., 2019) has tried to design “environment-independent” features in the first place.

This paper asks two questions. First, can the widely cited “breaks when the environment changes” phenomenon actually be measured on our experimental harness. Second, if it can, how much of that loss can be recovered by collecting a small amount of labeled target-domain data and adding it to the training set, i.e. by giving the installer a short on-site calibration step. The second question matters more operationally: re-labeling at camera-system scale every time a new room is added is unrealistic, but an installer spending a few minutes repeating a handful of prescribed motions is not. A measured answer to “how many repetitions is enough” is the goal of this paper.

3 Method: Domain Splits and Zero-Shot vs. Few-Shot Calibration

The experimental skeleton evaluates a source-trained classifier on a target domain under three conditions. Zero-shot uses no target-domain labels at all: the classifier trained on the source is applied to the target as is.

Few-shot calibration draws N examples per class (N in $\{1, 5, 10, 20, 50\}$) from a portion of the target domain set aside for calibration, adds them to the source training set, and retrains. The upper bound trains on the target-domain calibration data alone, with no source data at all, to measure how far a classifier could go if it fully committed to a target-specific model using as much target data as this protocol ever provides.

To keep the measurement fair, each repeat (12 repeats total, stratified random resampling) splits the target domain into a 70% calibration pool and a 30% held-out evaluation fold. The evaluation fold is never used for training under any condition. Zero-shot, few-shot, and the upper bound are all scored on the same evaluation fold within a repeat, so the three numbers are directly comparable. Few-shot draws always come from the calibration pool; when a class has fewer than N examples left in the pool, we take however many are available and log the actual count honestly (this capping only occurred in the official-split experiment described below, not in this paper’s main pseudo-session experiment). The feature scaler (StandardScaler) is always fit on that experiment’s source data only and merely applied, never refit, to the target domain. Refitting normalization on target statistics would itself be a form of domain adaptation, and the point of this experiment is to measure, in isolation, how much is recovered purely by adding a handful of labels.

The classifier reuses, unmodified, the 158-dimensional classical CSI feature set from the classical WiFi-CSI activity-sensing pipeline (per-subcarrier amplitude mean/variance/skewness/kurtosis, inter-antenna amplitude-difference variance, STFT band energy of the window’s dominant principal component, Hilbert envelope statistics, autocorrelation, spectral and histogram entropy) together with a LightGBM multiclass classifier. What this paper tests is not a new feature set or model but how the same pipeline behaves when the domain shifts, and how much a handful of labels fixes that, so feature extraction was reused rather than reimplemented. Models retrained per repeat (few-shot, upper bound) use 150 trees and 15 leaves, lighter than the original activity-classification setting (300 trees, 31 leaves), because the total repeat count (12 repeats times 3 domains times 6 conditions) is large; a preliminary check confirmed this reduction does not materially change accuracy.

4 Data and Setup: What Split Was Actually Available

This is the section where honesty matters most. UT-HAR is a public CSI activity dataset collected by Yousefi et al. (2017) with an Intel 5300 NIC (3 antennas by 30 subcarriers = 90 amplitude channels), and this experiment uses the train (3977)/val (496)/test (500) split exactly as redistributed by the SenseFi benchmark (Yang et al., 2023). The problem is that this redistribution publishes no room number, subject ID, or recording date/time. In other words, there is no information anywhere in the files about which room, which person, or which day a given window came from. Under these conditions, constructing a verified cross-room, cross-subject, or cross-day split is not possible, and this paper does not claim to have made one.

The first thing we tried was the only grouping variable that does exist in the data: the official train/val/test boundary, used as a domain-shift proxy. We trained on train and applied the classifier zero-shot to val and to test. The result was completely different from what we expected: 94.91% accuracy on val and 94.67% on test, essentially matching, within one standard deviation, the pipeline’s original in-split 7-way accuracy of 95.2%. In other words, crossing the official split boundary barely moved accuracy at all. This result is itself important information. The SenseFi paper states, in its own words, that segmenting with a sliding window “inevitably” causes repeated data among samples, and our zero-shot result reproduces exactly that caveat as a measurement: there is no evidence that val and test were collected under physically different conditions from train; if anything they behave as if they were cut out of the same continuous recording. Drawing a few-shot recovery curve on top of this split would be meaningless, because there is essentially no gap to recover (Section 4 shows the recovery figures bouncing between -6% and +2%, i.e. noise around zero). Interestingly, the “upper bound” condition (trained on target-only data) scored lower than zero-shot (76.73% on val, 75.89% on test), but this is not evidence of a real domain gap; it is a plain sample-size effect from training on only about 350 examples instead of nearly 4000. We are explicit that this number should not be read as “few-shot calibration makes things worse.”

Once it was clear the published metadata could not support a real domain-shift measurement, two options remained: abandon the domain-shift experiment and report only the negative finding, or construct an honestly labeled proxy axis ourselves. We chose the latter. Pooling train, val, and test into 4973 windows, we computed each window’s per-channel mean amplitude (90 dimensions, computed directly from the raw signal, separate from the 158-dimensional classifier feature set) and applied k-means clustering ($k=2$). Channel-mean amplitude is treated in the CSI literature as a physical proxy for receiver gain and multipath, so windows that end up in

the same cluster are more likely to share a physical recording condition, and windows in different clusters are more likely not to. The larger cluster contained 3624 windows and the smaller 1349, and both clusters covered all 7 activity classes (per-class counts in the smaller cluster range from 90 for the sit-down/stand-up pair to 447 for pickup). We designated the larger cluster as source and the smaller as target.

We do not hide the limits of this construction. It is not a verified room, subject, or day label, it is a cluster inferred by unsupervised clustering. Whether the two clusters actually correspond to different rooms, different subjects, different recording sessions, or some combination cannot be determined from this data alone. Nor is the clustering fully independent of the classification feature set, since several of the 158 classifier features are themselves aggregates of channel mean. What supports treating this as a genuine shift, rather than an artifact we manufactured, is the shape of the result itself: zero-shot accuracy drops sharply, and it climbs smoothly and monotonically as more target-domain data is added, a pattern that random noise does not produce. We refer to this proxy as “pseudo-session” throughout, and never as cross-room or cross-subject.

5 Results

The table below reports the mean and standard deviation over 12 repeats (fixed random seeds, stratified re-sampling each time). “Recovery” is (few-shot accuracy minus zero-shot accuracy) divided by (upper-bound accuracy minus zero-shot accuracy), times 100, i.e. the share of the zero-shot-to-upper-bound gap that few-shot calibration recovers.

Condition	pseudo-session (constructed shift)	official test (control)
Zero-shot accuracy	55.08% (SD 1.82pp)	94.67% (SD 1.72pp)
N=1 accuracy / recovery	57.16% / 5.6%	94.56% / 0.6%
N=5 accuracy / recovery	64.55% / 25.4%	94.67% / -0.0%
N=10 accuracy / recovery	70.62% / 41.7%	95.17% / -2.7%
N=20 accuracy / recovery	78.23% / 62.1%	95.06% / -2.1%
N=50 accuracy / recovery	84.92% / 80.0%	95.50% / -4.4%
Upper bound (target-only)	92.37% (SD 1.48pp)	75.89% (SD 2.61pp)
Zero-shot-to-upper-bound gap	37.29pp	-18.78pp (sample-size artifact)

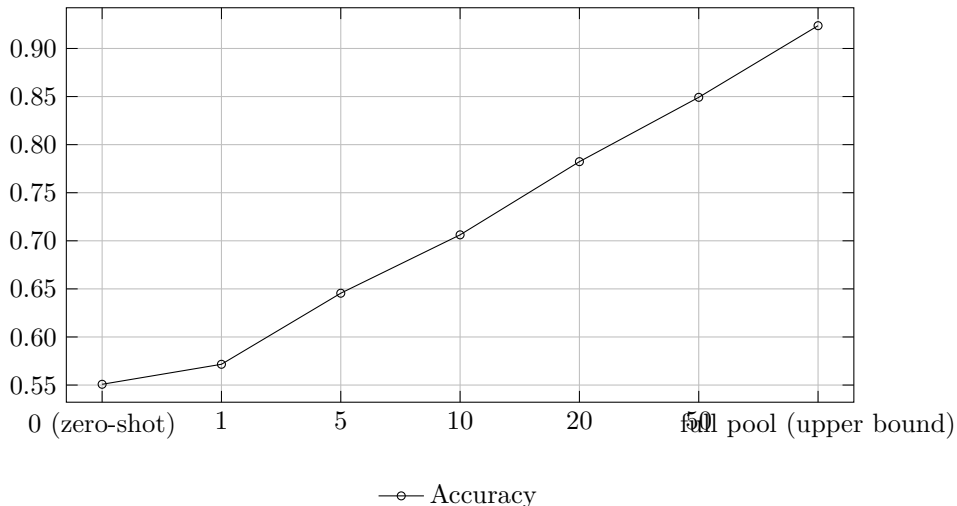


Figure 1: Few-shot accuracy recovery curve

The curve is smooth and monotonically increasing. Adding just 1 example per class raises accuracy by 2.08 percentage points; 5 examples raise it by 9.47 points; 10 by 15.54 points; 20 by 23.15 points; 50 by 29.84 points. Half of the total gap (18.65 points) is crossed somewhere between 10 and 20 shots per class, around 14 shots by

linear interpolation. Adding 50 examples per class, 350 labeled target-domain examples in total (only 9.7% of the 3624-example source training set), recovers 80% of the gap and brings accuracy to 84.92%. The remaining 20% of the gap (7.45 points) is not closed within the shot range tested here. Macro-F1 tells a slightly sharper story: zero-shot macro-F1 is only 43.68%, well below accuracy, meaning the minority classes (90-106 examples in the small cluster) are hit especially hard with no calibration. Macro-F1 rises to 82.19% at $N=50$, closing most of the gap to accuracy (84.92%), which suggests the small amount of calibration data lifts minority-class recognition along with the majority classes, not just the majority classes alone.

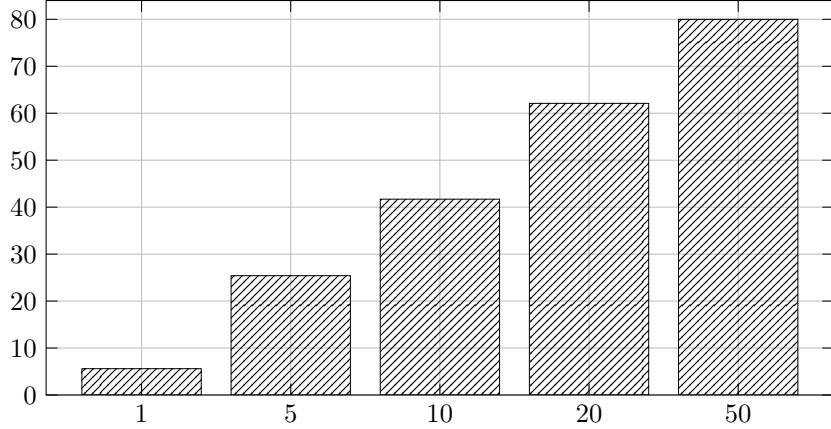


Figure 2: Recovery rate per shot count

The official-split control tells a completely different story. Because its zero-shot accuracy (94.67%) already sits near the in-split baseline, adding a handful of examples per class moves recovery between -4.4% and +0.6% with no clear direction. This does not mean few-shot calibration “does not work”; it means there was essentially no gap left to recover in the first place. As explained in Section 3, the lower accuracy of the upper-bound condition (75.89%) is a sample-size artifact from training on roughly 350 examples, not evidence of a real domain difference. We place this control next to the pseudo-session result specifically to show how sensitive a domain-shift measurement is to the choice of split, using the identical pipeline, model, and repeat protocol throughout.

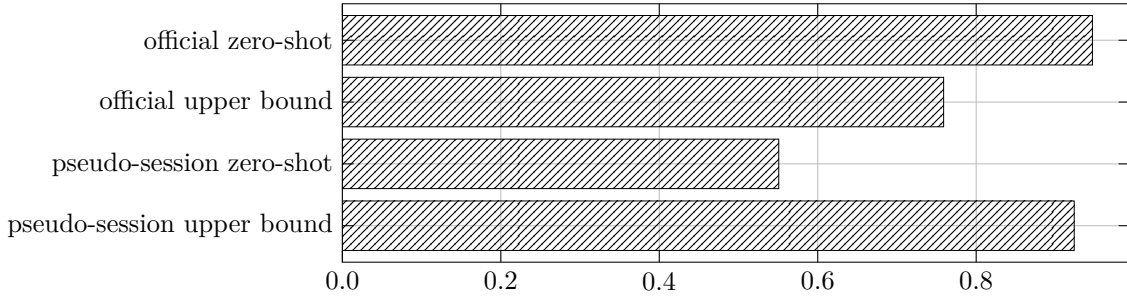


Figure 3: Split comparison: zero-shot and upper bound

The third chart compresses this paper’s methodological point into one figure. The official split shows zero-shot and the upper bound at almost the same height (with the upper bound even lower, from the sample-size effect), visually confirming there is no domain shift to measure there. The constructed split shows zero-shot dropping sharply and the upper bound sitting far above it, i.e. an axis that actually has something to measure. This contrast is the most direct answer to why this paper abandoned its originally planned split and built a new one instead.

6 Limitations

The domain shift reported here is not a verified cross-room, cross-subject, or cross-day split. UT-HAR’s SenseFi redistribution publishes no room, subject, or timestamp metadata, and we have not hidden that absence. The pseudo-session split is derived from unsupervised clustering on per-channel mean amplitude, and whether the two clusters actually reflect different rooms, different subjects, or an AGC reset from equipment being restarted cannot be determined from this data alone. The shape of the result, a smooth monotonic recovery curve, is circumstantial evidence that the clustering captured real structure rather than noise, not proof of it.

Second, the channel-mean amplitude used for clustering is not fully independent of the 158-dimensional classifier feature set, since several of those features are themselves aggregate statistics of the channel mean. We cannot fully rule out that part of the observed gap comes from the clustering criterion and the classification features partially overlapping rather than from a genuine environmental difference. That said, both clusters contain the full set of 7 activity classes, and class imbalance alone is unlikely to explain a drop to 55% zero-shot accuracy, so we judge this concern insufficient to invalidate the overall result.

Third, this experiment uses a single public dataset, a single classification pipeline (classical features plus LightGBM), and a single clustering method (k-means with $k=2$). A different dataset, more clusters, or a deep-learning domain-adaptation method could change the shape of the recovery curve or the number of shots required. This paper does not claim a universal constant for “how many shots are enough”; it reports one measured case for this specific pipeline on this specific data.

Fourth, the minority classes (90-106 examples in the smaller cluster) have enough repeats to average out but not enough to decompose per-class recovery speed separately, which this paper does not attempt.

Fifth, the “upper bound” is only the accuracy achievable from target-domain data alone; it is not a true theoretical ceiling. Collecting more target-domain data, or applying a more sophisticated domain-adaptation technique (feature alignment, adversarial training, etc.) that uses source and target together, could raise this ceiling further.

7 Reproduction

Data comes from UT-HAR (Yousefi et al., 2017) as redistributed by the SenseFi benchmark (Yang et al., 2023): three numpy arrays for train/val/test (X and y each), 250-step by 90-channel windows (3 antennas by 30 subcarriers), 7 activity classes (bed/fall/pickup/run/sitdown/standup/walk, alphabetical-order convention).

Feature extraction reuses, unmodified, the 158-dimensional handcrafted feature function from the original classical CSI activity-classification script (per-subcarrier mean/variance/skewness/kurtosis, whole-channel aggregate statistics, inter-antenna amplitude-difference variance, principal-component STFT band energy, Hilbert envelope statistics, short-time energy peakiness, autocorrelation, spectral and histogram entropy, global energy indicators). Features are computed per window independently, so no split introduces cross-sample leakage.

The official-split experiment trains a LightGBM multiclass classifier (150 trees, 15 leaves, learning rate 0.09) on the full train split (3977 windows) and applies it to val (496) and test (500) separately. The constructed-split experiment pools train, val, and test into 4973 windows, standardizes each window’s 90-dimensional channel-mean amplitude, applies k-means ($k=2$, seed 0, `n_init=10`), and designates the larger cluster (3624) as source and the smaller (1349) as target. Both experiments split the target domain into a stratified 70% calibration pool and 30% evaluation fold on every repeat (12 repeats, seeds 1000 through 1011), and score zero-shot, few-shot (1/5/10/20/50 shots per class), and the upper bound on the same evaluation fold. The feature scaler is fit on each experiment’s source data only.

The full pipeline runs on Python (numpy/scipy/scikit-learn/LightGBM), CPU-only, with no GPU or cloud resources required. Total run time, including feature extraction, is about 5 minutes for 12 repeats across the 3 domains.

8 References

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